Guidance and Control of Unmanned Autonomous Vehicles

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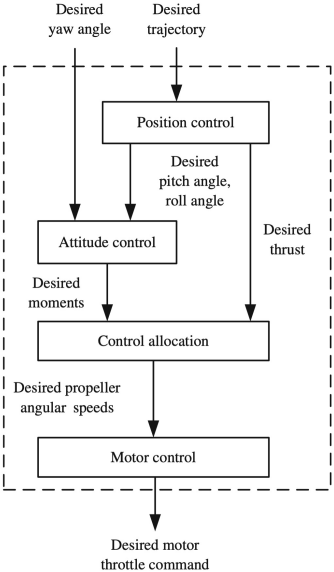
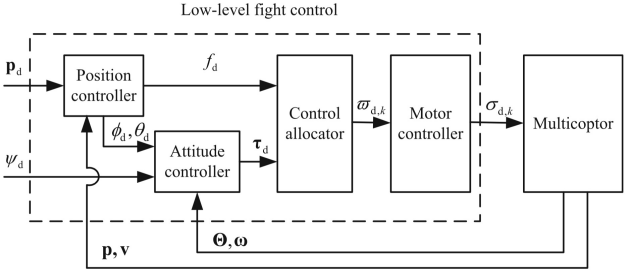
Quadrotor: Low Level Control

- **Unstable multicopters**: Stabilize using feedback control
- Is it enough to make it stable?
- No, What else is required then?
- A good control performance (such as fast response, no overshoot and robustness against uncertainties) required.
- How to control the multirotors?
- **Position control**: Compute ϕ_d, θ_d, f_d according to desired position $\boldsymbol{p}_d$
- **Attitude control**: Compute desired moments according to desired attitude angles ϕ_d, θ_d, ψ_d.
- **Control allocation**: Allocate desired propeller angular speeds to motors to obtain desired thrust and moments
- **Motor control**: Compute desired throttle command of each motor as per angular speed

Quadrotor: Low Level Control



Quadrotor: Low Level Control



- **Multicopters**: Underactuated, Why?
- 6 outputs (position and attitude) but only 4 independent inputs (thrust and moments)
- Multicopter can only track four desired commands ($\boldsymbol{p}_d \in \mathbb{R}^3, \psi_d \in \mathbb{R}$)
- Remaining variables ($\phi_d, \theta_d \in \mathbb{R}$) determined by desired commands $\boldsymbol{p}_d, \psi_d$.

Quadrotor: Low Level Control

- Nonlinear model for the quadrotor

$$
\begin{aligned}
{}^{e}\dot{\boldsymbol{p}} &= {}^{e}\boldsymbol{v}, \quad {}^{e}\dot{\boldsymbol{v}} = g\boldsymbol{e}_3 - \frac{f}{m}\boldsymbol{\mathcal{R}}_b^e \boldsymbol{e}_3 \\
\dot{\boldsymbol{\Theta}} &= \boldsymbol{W}\ {}^{b}\boldsymbol{\omega}, \quad \boldsymbol{J}\ {}^{b}\dot{\boldsymbol{\omega}} = -{}^{b}\boldsymbol{\omega} \times \boldsymbol{J}\,{}^{b}\boldsymbol{\omega} + \boldsymbol{G}_a + \boldsymbol{\tau}; \quad (\dot{\boldsymbol{\mathcal{R}}} = \boldsymbol{\mathcal{R}}[{}^{v}\boldsymbol{\omega}]^{\times})
\end{aligned}
$$

- **Simplified model for the quadrotor**

$$
{}^{e}\dot{\boldsymbol{p}} = {}^{e}\boldsymbol{v}, \quad {}^{e}\dot{\boldsymbol{v}} = g\boldsymbol{e}_3 - \frac{f}{m}\boldsymbol{\mathcal{R}}_b^e \boldsymbol{e}_3, \quad \dot{\boldsymbol{\Theta}} = \boldsymbol{\mathcal{W}}\ {}^{b}\boldsymbol{\omega}, \quad \boldsymbol{J}\ {}^{b}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}
$$

- Component-wise equations governing the quadrotor's position

$$
\begin{aligned}
\ddot{p}_{x_e} &= -\frac{f}{m}\left(\sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi\right) \\
\ddot{p}_{y_e} &= -\frac{f}{m}\left(-\cos\psi\sin\phi + \sin\psi\sin\theta\cos\phi\right) \\
\ddot{p}_{z_e} &= g - \frac{f}{m}\cos\theta\cos\phi
\end{aligned}
$$

Quadrotor: Linear Model Simplification

- Quadrotor is underactuated, strongly coupled, and high dimensional.
- Simplification based on the flight characteristics.
- Small pitch and roll angles along with thrust being equal to weight

$$\sin\phi \approx \phi, \quad \cos\phi \approx 1, \quad \sin\theta \approx \theta, \quad \cos\theta \approx 1, \quad f \approx mg$$

$$\boldsymbol{\mathcal{W}} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix} \approx \boldsymbol{I}, \quad \boldsymbol{\mathcal{R}}_b^e \boldsymbol{e}_3 \approx \begin{bmatrix} \theta\cos\psi + \phi\sin\psi \\ \theta\sin\psi - \phi\cos\psi \\ 1 \end{bmatrix}$$

- **Decoupled channels**
  - Horizontal position channel model
  - Altitude channel model
  - Attitude model

Quadrotor: Horizontal Position Channel

- Under small angle assumptions, quadrotor horizontal position model

$$\boxed{\dot{\boldsymbol{p}}_h = \boldsymbol{v}_h, \quad \dot{\boldsymbol{v}}_h = -g\boldsymbol{\mathcal{A}}_\psi \Theta_h}$$

$$\boldsymbol{p}_h = \begin{bmatrix} p_x \\ p_y \end{bmatrix}, \boldsymbol{\Theta}_h = \begin{bmatrix} \phi \\ \theta \end{bmatrix}$$

$$\boldsymbol{\mathcal{R}}_\psi = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}, \quad \boldsymbol{\mathcal{A}}_\psi = \boldsymbol{\mathcal{R}}_\psi \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

- Input: $-g\boldsymbol{A}_\psi \boldsymbol{\Theta}_h$
- Output: $\boldsymbol{p}_h$
- Linear model
- Second-order integrator model

Quadrotor: Vertical Position Channel

- Under small angle assumption, quadrotor altitude model

$$\dot{p}_z = v_z, \qquad \dot{v}_z = g - \frac{f}{m}$$

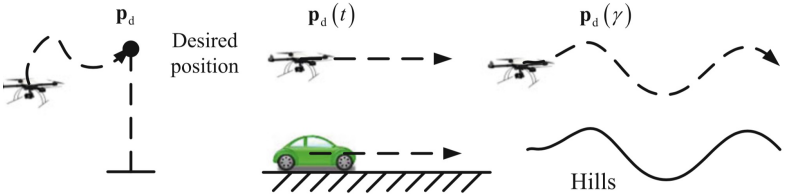
- $\left(g - \frac{f}{m}\right)$ is not a higher-order infinitely small term, cannot be ignored.
- Again a linear model, (**second order integrator model**)
- Attitude linear model

$$\dot{\boldsymbol{\Theta}} = \boldsymbol{\omega}, \quad \boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}$$

- States may not always be available and thus must be estimated.
- Design of controller using the **separation principle**

Quadrotor: Position Control

- **Position control**: Set-point control, trajectory tracking, and path following



(a) Set-point control (b) Trajectory tracking (c) Path following

- **Set-point control**: Desired position $\boldsymbol{p}_d \in \mathbb{R}^3$ is a constant point.

$$\|\boldsymbol{p} - \boldsymbol{p}_d\| \to 0, \quad \text{or} \quad \|\boldsymbol{p} - \boldsymbol{p}_d\| \to \mathcal{B}\left(\mathbf{0}_{3\times 1}, \delta\right) \text{ as } t \to \infty$$

$$\mathcal{B}\left(\boldsymbol{o}_{3\times 1}, \delta\right) \triangleq \{\boldsymbol{\zeta} \in \mathbb{R}^3 | \|\boldsymbol{\zeta} - \boldsymbol{o}\| \leq \delta\}, \ \delta \in \mathbb{R}_+ \cup \{0\}$$

- Multicopter can fly to the given position, regardless of the trajectory traversed.

Quadrotor: Position Control

- **Trajectory tracking**: $\boldsymbol{p}_d : [0, \infty) \in \mathbb{R}^3$ is a time-dependent desired trajectory.

$$\|\boldsymbol{p}(t) - \boldsymbol{p}_d(t)\| \to 0, \quad \text{or} \quad \|\boldsymbol{p}(t) - \boldsymbol{p}_d(t)\| \to \mathcal{B}\left(\mathbf{0}_{3\times 1}, \delta\right) \text{ as } t \to \infty$$

- **Example**: Multicopter required to track a given trajectory over a car.
- **Path following**: Desired position $\boldsymbol{p}_d(\gamma(t)) \in \mathbb{R}^3$ is path directly determined by the parameter γ, rather than the time t.

$$\|\boldsymbol{p}(t) - \boldsymbol{p}_d(\gamma(t))\| \to 0, \quad \text{or} \quad \|\boldsymbol{p}(t) - \boldsymbol{p}_d(\gamma(t))\| \to \mathcal{B}\left(\mathbf{0}_{3\times 1}, \delta\right) \text{ as } t \to \infty$$

- **Example**: Multicopter required to fly along a profile over a hill
- **Difference in trajectory tracking and path following**: Dependence of the trajectories or paths on time
- **Trajectory tracking problem**: Time-dependent desired trajectory
- **Path following problem**: Curve independent of time

Quadrotor: Position Control

- **Path following**: Three-dimensional (3D) tracking.
- **Trajectory tracking**: Four-dimensional (4D) tracking with the time as a new dimension.
- When $\gamma(t) = t$, path following problem $\mapsto$ a trajectory tracking problem.
- If $\boldsymbol{p}_d(t) =$ constant, trajectory tracking problem $\mapsto$ a set-point control problem.
- **Set-point control** problem: A special case of **trajectory tracking** problems.
- **Trajectory tracking** problem: A special case of **path following** problems.
- Transformation of path following problems to trajectory tracking problems
- Desired attitude control problems
  - Euler angles
  - Rotation matrix

Quadrotor: Euler angles based Approach, Horizontal Position Channel

- **Horizontal position control**: Desired attitude $\boldsymbol{\Theta}_{hd} = (\phi_d, \theta_d)^\top$ s.t.

$$\lim_{t\to\infty} \|\boldsymbol{e}_{p_h}(t)\| = 0, \quad \boldsymbol{e}_{p_h}(t) = \boldsymbol{p}_h(t) - \boldsymbol{p}_{hd}(t)$$

- How to design controller now? Expected error dynamics

$$\ddot{\boldsymbol{e}}_{p_h} + \boldsymbol{K}_{p_{hd}}\dot{\boldsymbol{e}}_{p_h} + \boldsymbol{K}_{p_{hp}}\boldsymbol{e}_{p_h} = \boldsymbol{0} \implies \boxed{\ddot{\boldsymbol{p}}_h = \ddot{\boldsymbol{p}}_{hd} - \boldsymbol{K}_{p_{hd}}\dot{\boldsymbol{e}}_{p_h} - \boldsymbol{K}_{p_{hp}}\boldsymbol{e}_{p_h}}$$

- As $\ddot{\boldsymbol{p}}_h = -g\boldsymbol{A}_\psi\boldsymbol{\Theta}_h$, we have $-g\boldsymbol{A}_\psi\boldsymbol{\Theta}_{hd} = \ddot{\boldsymbol{p}}_{hd} - \boldsymbol{K}_{p_{hd}}\dot{\boldsymbol{e}}_{p_h} - \boldsymbol{K}_{p_{hp}}\boldsymbol{e}_{p_h}$

$$\boxed{\implies \boldsymbol{\Theta}_{hd} = -g^{-1}\boldsymbol{A}_\psi^{-1}\left(\ddot{\boldsymbol{p}}_{hd} - \boldsymbol{K}_{p_{hd}}\dot{\boldsymbol{e}}_{p_h} - \boldsymbol{K}_{p_{hp}}\boldsymbol{e}_{p_h}\right)}$$

- If $\boxed{\boldsymbol{\Theta}_h = \boldsymbol{\Theta}_{hd}}$ then $\lim_{t\to\infty}\|\boldsymbol{e}_{p_h}(t)\| = 0$
- For a set-point control problem, $\ddot{\boldsymbol{p}}_{hd} = \dot{\boldsymbol{p}}_{hd} = \boldsymbol{0}$ then

$$\boxed{\boldsymbol{\Theta}_{hd} = -g^{-1}\boldsymbol{A}_\psi^{-1}\left[-\boldsymbol{K}_{p_{hd}}\dot{\boldsymbol{p}}_h - \boldsymbol{K}_{p_{hp}}(\boldsymbol{p}_h - \boldsymbol{p}_{hd})\right]}$$

Quadrotor: Euler angles based Approach, Altitude Channel

- **Altitude Channel**: Control objective: Choose desired thrust f_d such that

$$\lim_{t\to\infty} |e_{p_z}(t)| = 0, \quad e_{p_z}(t) = p_z - p_{zd}$$

- Control design? Expected error dynamics

$$\ddot{e}_{p_z} + K_{p_{zd}}\dot{e}_{p_z} + K_{p_{zp}}e_{p_z} = 0 \implies \boxed{\ddot{p}_z = \ddot{p}_{zd} - K_{p_{zd}}\dot{e}_{p_z} - K_{p_{zp}}e_{p_z}}$$

- How to proceed now? As $\ddot{p}_z = g - \dfrac{f_d}{m}$,

$$g - \frac{f_d}{m} = \ddot{p}_{zd} - K_{p_{zd}}\dot{e}_{p_z} - K_{p_{zp}}e_{p_z} \implies \boxed{f_d = mg - m\left[\ddot{p}_{zd} - K_{p_{zd}}\dot{e}_{p_z} - K_{p_{zp}}e_{p_z}\right]}$$

- If $\boxed{f = f_d}$ then $\lim_{t\to\infty} \|e_{p_z}(t)\| = 0$
- For a set-point control problem, $\ddot{p}_{zd} = \dot{p}_{zd} = 0$ then

$$\boxed{f_d = mg - m\left[-K_{p_{zd}}\dot{p}_z - K_{p_{zp}}(p_z - p_{zd})\right]}$$

Quadrotor: Euler angles based Approach, Horizontal Position Channel, PID

- To satisfy $\lim_{t\to\infty} \|\boldsymbol{e}_{p_h}(t)\| = 0$ using $\dot{\boldsymbol{p}}_h = \boldsymbol{v}_h$, desired value of $\boldsymbol{v}_h$

$$\boxed{\boldsymbol{v}_{hd} = \boldsymbol{K}_{p_h}\left(\boldsymbol{p}_{hd} - \boldsymbol{p}_h\right)}$$

- With assumption of $\dot{\boldsymbol{p}}_{hd} = \boldsymbol{0}$ and $\boldsymbol{e}_{v_h} \triangleq \boldsymbol{v}_h - \boldsymbol{v}_{hd}$

$$\lim_{t\to\infty} \|\boldsymbol{e}_{v_h}(t)\| = 0$$

$$\implies \lim_{t\to\infty} \|\boldsymbol{e}_{p_h}(t)\| = 0$$

- As $\dot{\boldsymbol{v}}_h = -g\boldsymbol{A}_\psi \boldsymbol{\Theta}_h$, we choose the PID controller as

$$-g\boldsymbol{A}_\psi \boldsymbol{\Theta}_{hd} = -\boldsymbol{K}_{v_{hp}}\boldsymbol{e}_{v_h} - \boldsymbol{K}_{v_{hi}}\int \boldsymbol{e}_{v_h} - \boldsymbol{K}_{v_{hd}}\dot{\boldsymbol{e}}_{v_h}$$

- With assumption of $\dot{\boldsymbol{v}}_{hd} = \boldsymbol{0}$,

$$\lim_{t\to\infty} \|\boldsymbol{\Theta}_h(t) - \boldsymbol{\Theta}_{hd}(t)\| = 0 \implies \lim_{t\to\infty} \|\boldsymbol{e}_{v_h}(t)\| = 0$$

- Desired attitude angle

$$\boxed{\boldsymbol{\Theta}_{hd} = g^{-1}\boldsymbol{A}_\psi^{-1}\left[\boldsymbol{K}_{v_{hp}}\boldsymbol{e}_{v_h} + \boldsymbol{K}_{v_{hi}}\int \boldsymbol{e}_{v_h} + \boldsymbol{K}_{v_{hd}}\dot{\boldsymbol{e}}_{v_h}\right]}$$

Quadrotor: Euler angles based Approach, Horizontal Position Channel, PID

- What do we expect from the attitude controller of the quadrotor?
- **Attitude angle controller objective**: To ensure that

$$\lim_{t\to\infty} \|\boldsymbol{\Theta}_h(t) - \boldsymbol{\Theta}_{hd}(t)\| = 0$$

- As a result, $\lim_{t\to\infty} \|\boldsymbol{e}_{p_h}\| = 0$
- For a set point control problem, $\ddot{\boldsymbol{p}}_{hd} = \dot{\boldsymbol{p}}_{hd} = 0$
- $\boldsymbol{K}_{v_{hd}}\dot{\boldsymbol{e}}_{v_h}$ can be omitted to avoid noise by the differential mechanism.
- Why only P-controller for position but the PID for velocity?
- First one is based on kinematic model (uncertainty free) but second one is using dynamic model, subject to uncertainties.
- PID can compensate in a better manner for the uncertainties in the systems.

Quadrotor: Euler angles based Approach, Altitude Channel, PID

- To satisfy $\lim_{t\to\infty} |e_{p_z}(t)| = 0$ using $\dot{p}_z = v_z$, $\boxed{v_{zd} = K_{p_z}\left(p_{zd} - p_z\right)}$
- With assumption of $\dot{p}_z = 0$ and $e_{v_z} = v_z - v_{zd}$,

$$\lim_{t\to\infty} |\boldsymbol{e}_{v_z}(t)| = 0 \implies \lim_{t\to\infty} |\boldsymbol{e}_{p_z}(t)| = 0$$

- As $\dot{v}_h = g - \dfrac{f}{m}$, we choose PID controller, given by

$$g - \frac{f_d}{m} = -K_{v_{zp}} e_{v_z} - K_{v_{zi}} \int e_{v_z} - K_{v_{zd}} \dot{e}_{v_z}$$

- With $\dot{v}_{zd} = 0$, $\lim_{t\to\infty} |f(t) - f_d(t)| = 0 \implies \lim_{t\to\infty} |e_{v_z}(t)| = 0$
- Desired thrust: $\boxed{f_d = m\left(g + K_{v_{zp}} e_{v_z} + K_{v_{zi}} \int e_{v_z} + K_{v_{zd}} \dot{e}_{v_z}\right)}$
- For set-point problem, $\dot{p}_{zd} = \ddot{p}_{zd} = 0$ and the derivative term $K_{v_{zd}} \dot{e}_{v_z}$ can be omitted to reduce the noise in input.

References

References

- Quan Quan, "Introduction to Multicopter Design and Control", Springer, 2017.
- Moussa Labbadi, "Advanced Robust Nonlinear Control Approaches for Quadrotor Unmanned Aerial Vehicle", 2021.

Thank you for your attention !!!